Supporting Information for
BioInteract: Interpretable Drug–Target Interaction Prediction via Residue-Level Cross-Attention with Biological Prior Knowledge

This document reports values recoverable from the released Davis records, configuration files, saved result files, source code, and the fixed BindingDB external-validation manifest included with this revision. "Classifier score" denotes the sigmoid output for the binary high-affinity class; it is not an experimental affinity. Model-native attention and Grad-CAM values are attribution scores, not contact assignments.

Model configuration and input features

Table S1: Model architecture confirmed from the checkpoint-embedded model configurations.

Component	Archived setting
Drug encoder	GINE; 3 layers; hidden dimension 256; dropout 0.20
Atom and bond input	52 node features; 16 edge features
Protein language model	`esm2_t30_150M_UR50D`; 640-dimensional residues
Protein projection	256-dimensional projection; 32-dimensional domain-label embedding
Cross-attention	8 heads; attention dimension 256; dropout 0.10
Pooling	Gated
Prediction head	Hidden dimensions 256 and 128; dropout 0.30; classification task

Table S2: Protein-residue input channels and their availability.

Channel	Archived implementation and limitation
ESM-2	Frozen 640-dimensional residue embeddings; proteins are truncated at 1,200 residues.
Physicochemical vector	Four values per residue: hydrophobicity, normalized molecular weight, charge, and polarity.
Domain-label channel	The configuration reserves a 32-dimensional embedding. The label builder uses `NONE` when no per-protein JSON annotation is available.
Annotation availability	No `data/domain_annotations/*.json` files are present in the released package. Thus the reproducible Davis input uses the unknown label at every residue and is not curated Pfam/InterPro information.

Table S3: Training provenance for the historical entity-split checkpoints and the separately trained exact-sequence-grouped strict models.

Setting	Value
Historical entity-split scope	Model architecture only; the three historical entity-split checkpoints do not retain optimizer, scheduler, warmup, or loss state
Historical Random and Drug-ID logs	Final-run headers record weighted BCE, label smoothing 0.05, AdamW (10^{-4}, 10^{-5}), and cosine warmup
Historical Target-ID trace	Older `run_split.py` header; only loss and validation AUROC are logged, so optimizer, warmup, and smoothing are not recoverable from the artifact
Historical threshold protocol	A validation F1 threshold is frozen before each canonical test evaluation
Strict-model initialisation and data	Each exact-sequence-grouped model was newly initialised with seed 42 and trained only on its new Davis training partition, using the same preprocessing and architecture
Strict-model optimisation	Weighted BCE with 0.05 label smoothing; AdamW (learning rate 10^{-4}, weight decay 10^{-5}); batch size 64; gradient-norm clipping 1.0; five-epoch linear warm-up followed by cosine decay; maximum 100 epochs
Strict-model selection	Highest validation AUROC with patience 15; validation F1 threshold selected after loading that checkpoint and frozen for test evaluation
Strict cold-target artifact	`revision/analysis/sequence_grouped_cold_target.pt`; best epoch 24 of 39; validation AUROC 0.911568; threshold 0.837581; SHA-256 `8ea58e1f056b5e33c309787880a811c797393510c459d9442b5dbe67e632fe5c`
Strict cold-both artifact	`revision/analysis/sequence_grouped_cold_both.pt`; best epoch 6 of 21; validation AUROC 0.910630; threshold 0.935704; SHA-256 `8b7c0f374983a35163a7d3243c89732504a5a97ef4d10c557b2841a627873972`

Table S4: Computational environment documented during final reproducibility audit.

Item	Value
Python	3.10.14
PyTorch	2.4.0+cu121
PyTorch Geometric	2.6.1
RDKit	2025.3.6
`fair-esm`	2.0.0
Biopython and RapidFuzz	1.87 and 3.14.5
CUDA availability during audit	Available
Audit GPU	NVIDIA GeForce RTX 4080
Strict-split run record	`device: cuda` in `strict_split_metrics.json`
Historical checkpoint files	`best.pt`, `best_random.pt`, `best_cold_target.pt`, `best_cold_drug.pt`
Strict-model checkpoint files	`revision/analysis/sequence_grouped_cold_target.pt` and `revision/analysis/sequence_grouped_cold_both.pt` (hashes in Table S3)

Dataset, partitions, and metrics

Table S5: Canonical current-release Davis test-partition summaries. Values are from the full-precision checkpoint reconstruction, frozen validation thresholds, and the associated prediction files.

Partition	Test pairs	Positives	AUROC	AUPRC	F1	Precision	Recall
Random	6,011	276	0.904	0.560	0.565	0.602	0.533
Target-ID-held-out	5,984	250	0.930	0.525	0.534	0.517	0.552
Drug-ID-held-out	5,746	245	0.733	0.167	0.100	0.273	0.061

Table S6: Bootstrap intervals for the canonical current-release entity-based partitions. Each interval uses 600 ordinary pair-level bootstrap replicates sampled with replacement at the test-pair level, NumPy seed 42, and the 2.5th and 97.5th percentiles of valid replicate metrics. Replicates containing only one class are discarded. The main figure shows point estimates only.

Partition	AUROC 95% interval	AUPRC 95% interval
Random	[0.881, 0.924]	[0.498, 0.624]
Target-ID-held-out	[0.911, 0.948]	[0.454, 0.584]
Drug-ID-held-out	[0.703, 0.763]	[0.129, 0.211]

Table S7: Exact-sequence-grouped strict within-Davis evaluations from `strict_split_metrics.json`. The two models were trained from initialisation under the protocol in Table S3. Threshold-dependent quantities use the corresponding validation-set threshold; all strict-split entries are point estimates, not bootstrap confidence intervals.

Pair accounting

Protocol	Train pairs	Validation pairs	Test pairs	Test positives	Excluded cross-block pairs
Exact-sequence-grouped cold-target	21,080	2,992	5,984	208	0
Exact-sequence-grouped cold-both	15,190	264	1,144	38	13,458

Test metrics

Protocol	AUROC	AUPRC	Threshold	F1	Precision	Recall
Exact-sequence-grouped cold-target	0.873	0.383	0.838	0.468	0.517	0.428
Exact-sequence-grouped cold-both	0.552	0.038	0.936	0.000	0.000	0.000

For the cold-both protocol, drug identifiers and exact target-sequence groups are partitioned independently at 70/10/20. Only pairs in the corresponding train–train, validation–validation, and test–test blocks are retained; the remaining cross-block pairs are excluded. The cold-both validation block contains 10 positive pairs.

Reproducibility details for revision analyses

Identifier-partition similarity audits. Drug similarities use RDKit 2025.3.6 Morgan fingerprints (radius 2, 1,024 bits, `includeChirality=False`) and Tanimoto similarity. Protein similarities for the original Target-ID-held-out audit use RapidFuzz 3.14.5 `fuzz.ratio`/100. For sequences x, y, this is the normalised Indel similarity $1 - d_{\mathrm{Indel}}/(|x| + |y|)$, where substitutions are represented as one deletion plus one insertion.

Strict-test global alignment. Biopython 1.87 `Align.PairwiseAligner` is used in global mode with match = 1, mismatch = 0, gap-opening = -1, and gap-extension = -0.5. Full-length identity is the number of exact-residue matches divided by alignment length; the maximum across all training targets is retained for each held-out target.

Bootstrap intervals. The saved 95% intervals use ordinary, unstratified pair-level resampling with replacement, 600 requested replicates, and NumPy random seed 42. Each replicate contains the same number of test pairs as the corresponding test set; one-class replicates are skipped. Bounds are the 2.5th and 97.5th percentiles of the valid AUROC or AUPRC replicate distribution.

Grad-CAM. Hooks are attached to `drug_encoder.layers.2`, the third and final GINE message-passing layer. Gradients are taken from the pre-sigmoid positive-class logit, pooled across graph nodes to obtain channel weights, rectified, and then normalised separately within each molecular graph.

Attention sparsity. For pair q, residue scores $s_{qj} = \sum_i M_{qij}$ are normalised as $\widetilde{s}_{qj} = s_{qj}/\max_k s_{qk}$. The reported fraction is $|\{(q,j) : \widetilde{s}_{qj} < 0.1\}|/\sum_q L_q$. It uses all 1,506 positive Davis pairs with the archived `checkpoints/best.pt`, not a single test set. Every pair is normalised before the 1,301,380 valid residue scores are concatenated; thus original training, validation, and test memberships are all included and longer targets contribute more residue-level observations. The archived flat-score array is `BioInteract/results/figure_data/attention_distribution.npz`; it is the sole source of the attention-distribution figure and cumulative curve in the main text.

Residue-attribution summary.

Quantity	Value
Positive Davis pairs	1,506
Residue-score observations	1,301,380
Mean / standard deviation	0.004351 / 0.046243
Median	0.000272
95th / 99th percentile	0.003617 / 0.057193
Fraction below 0.1	0.992319
Attribution mass in top 1% / 5% scores	0.788528 / 0.904465

Exact-sequence-grouped target diagnostic controls. All values use the 5,984-pair strict target test set with 208 positives.

Control	AUROC	AUPRC
BioInteract	0.873	0.383
Drug-only Morgan logistic regression	0.764	0.164
Drug-promiscuity prior	0.764	0.163
Target-only constant prior	0.500	0.035
Label-shuffled drug-only control	0.407	0.029

Feature and input audits

Table S8: Atom and bond feature specification from `src/data/mol_graph.py`.

Level	Feature family	Dimensions
Atom	Atom type one-hot (20 choices including unknown)	20
Atom	Hybridization, formal charge, and hydrogen-count one-hot vectors	18
Atom	Degree, aromaticity, and ring membership	3
Atom	Ring-size membership for sizes 3–8	6
Atom	Donor/acceptor, Crippen logP and MR, Gasteiger charge	5
Atom total		52
Bond	Bond type, conjugation, ring membership, stereo, and direction	16

Table S9: Davis nominal ABL1 input audit. Every listed record has the same complete SHA-256 digest `cd6a78b82fede215cdb80a797f228098a41178fdde0753728f3781e6221fb1ee`; values in the last columns are the archived D0010 and D0017 affinities in nM. The nominal labels do not alter the model sequence input.

Target ID	Nominal name	Length	Matches ABL1	D0010	D0017
T0001	ABL1(E255K)	1167	Yes	0.047	0.047
T0002	ABL1(F317I)	1167	Yes	0.63	0.10
T0003	ABL1(F317I)p	1167	Yes	0.18	0.041
T0004	ABL1(F317L)	1167	Yes	0.11	0.032
T0005	ABL1(F317L)p	1167	Yes	0.029	0.019
T0006	ABL1(H396P)	1167	Yes	0.062	0.025
T0007	ABL1(H396P)p	1167	Yes	0.057	0.046
T0008	ABL1(M351T)	1167	Yes	0.037	0.016
T0009	ABL1(Q252H)	1167	Yes	0.086	0.037
T0010	ABL1(Q252H)p	1167	Yes	0.039	0.064
T0011	ABL1(T315I)	1167	Yes	21	890
T0012	ABL1(T315I)p	1167	Yes	3.6	120
T0013	ABL1(Y253F)	1167	Yes	0.036	0.058
T0014	ABL1	1167	Yes	0.12	0.029
T0015	ABL1p	1167	Yes	0.057	0.046

Table S10: Saved non-mutant EGFR–D0014 functional-group attribution values.

Functional group	Attribution
Carbonyl	0.431
Amide	0.389
Ether	0.136
Halogen	0.134
Amino	0.090
Aromatic ring	0.000
Heterocycle N	0.000

Table S11: Saved non-mutant BRAF–D0023 failure-mode summary. The archived affinity is positive at the study threshold, whereas the classifier score is low. The score and attributions are model outputs, not an explanation of binding.

Quantity	Value
High-affinity-class score	0.099
Archived Davis affinity (nM)	0.19
Aromatic-ring attribution	0.059
Hydroxyl attribution	0.000
Heterocycle-N attribution	0.000

Table S12: Trainable parameter count recovered from `interpretability_report.json`.

Quantity	Value
Total trainable parameters	2,442,083

Strict external BindingDB evaluation

Table S13 records the frozen external evaluation requested by the reviewers. The official curated-articles snapshot was handled independently of Davis, then filtered under the pre-specified exact-K_d and exact double-novel protocol. This is an external exact-entity test, not a claim of scaffold or remote-homology novelty, and no BindingDB label was used to retrain, calibrate, select the checkpoint, or select the threshold.

Table S13: Strict BindingDB external-validation provenance, curation, and frozen result. The source archive was `BindingDB_BindingDB_Articles_202607_tsv.zip` (SHA-256 `d2584d1519318d00ab5f46289da5ab3549affe732d598a5072f8777b6b3b5262`). Only finite exact K_d measurements in nM for valid single-chain proteins (maximum 1,200 residues) and valid ligand structures were considered. Exact canonical-isomeric-SMILES and full-normalised-sequence matches to every Davis ligand or target were excluded. Eligible measurements were grouped by canonical ligand–sequence pair; threshold-conflicting groups were removed at the pair level, and every remaining pair retained its median exact K_d. Confidence intervals are pair-stratified bootstrap 95% intervals from 600 replicates (seed 42).

Item	Value
Source snapshot	`BindingDB_BindingDB_Articles_202607` (official BindingDB curated articles TSV)
Source rows	93,712
Final positives / negatives	516 / 1,416
Unique ligands / targets	1,195 / 396
Sequential row-level exclusions	82,988 censored/non-exact K_d; 8,261 multi-chain; 139 invalid sequences; 8 invalid SMILES; 64 Davis-ligand matches; 94 Davis-sequence matches
Eligible measurements before pair reconciliation	2,158
Threshold-conflicting replicate groups	15 ligand–target pairs, comprising 44 measurement rows; conflicts are counted at the pair level
Threshold-consistent measurement rows	2,114
Median replicate aggregation	143 retained pairs had $\geq$ 2 consistent measurements; 182 repeated measurement rows were collapsed by the within-pair median K_d
Final external pairs	1,932
Frozen model and threshold	`checkpoints/best_random.pt`; 0.5959881544 (existing Davis random-validation threshold)
Protein features	Newly extracted `fair-esm` 2.0.0 `esm2_t30_150M_UR50D`; all 396 caches passed 640-dimensional and length checks
AUROC (95% CI)	0.5597 [0.5313, 0.5885]
AUPRC (95% CI)	0.3404 [0.3138, 0.3724]
Frozen-threshold F1 / precision / recall	0.0701 / 0.7308 / 0.0368
Two-run reproducibility	Identical per-pair probabilities; the complete SHA-256 is recorded in `bindingdb_external_manifest.json`
Interpretation	Limited external transfer; the result does not support a broad DTI-transfer claim and is not used for structural or attribution interpretation.